Chapter 12 Answers

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Exercise

Note: Smullyan omits saying what alphabet we should show Ax represents the set of all alternating strings, so I will assume he meant the alphabet $\{a, b\}$ (as this is the one he defines (E) with).
We are to show that Ax really represents the set of all alternating strings over the alphabet $\{a, b\}$.

Proof. On the one hand, assume that Ax is provable. In the base case where x a, b, ab or ba, these are all alternating by definition. So let us make an induction assumption that for all Ax provable with x of length n, x is alternating. Well, the only formulas of length $n+1$ for which Ax could be provable would be of the form $Axab$ where Axa is provable, or of the form $Axba$ where Axb is provable. In both cases, we have Axa, Axb having xa and xb both being alternating by the assumption, thus meaning it is obvious that $Axab, Axba$ would both be alternating, with respect to the provable sentences xa, xb.

Conversley, we wish to show that all alternating strings x in the alphabet $\{a, b\}$ are such that Ax is provable. The base cases of lengths 1 and 2 are given by the first four facts of the system. Now, make an induction assumption that all alternating strings s from the alphabet a, b of degree n are such that Ax is provable. Thus s must either be of the form xa or xb for some x. Hence, because s is either xa or xb, either $Axab$ or $Axba$ are provable in (E), respectively. Also, the only ay to make an alternating string starting from some string of the from xa or xb is from the string xab or xba. Hence, all of the alternating strings of degree $n+1$ are such that Ax is provable, hence proving that all alternating strings from the alphabet $\{a, b\}$ are such that Ax is provable, thus completing the proof. □

Problem 1

We are considering two EFS's E_1, E_2 over the alphabet K in which the sets W_1, W_2 are represented, respectively. Well, we consider all of those predicates common to both systems. We then replace these common predicates in E_2 by other symbols of the predicates, and keep all of the predicates in E_1. Then, we consider the EFS E which is the combination of both of the axiom schemes of E_1 and the altered system of E_2. Because all the axiom schemes are the same, the only thing that could have made W_1, W_2 no longer representable is if 'extra' axiom schemes using the same predicates as the ones that represent them were added. However, this was guaranteed not to happen as all of the possible 'added' predicates would be due to predicates in common to E_1, E_2, which we changed when we redefined E_2. Thus, W_1 and W_2 are still both representable.

Problem 2

We suppose W_1, W_2 are representable over K. Hence, by Problem 1, there is some (E) over K such that both W_1, W_2 are representable. So, we simply add to (E) the predicate P with the following axioms:
$P_1x \to Px$
$P_2x \to Px$ for P_1, P_2 the predicates that represent W_1, W_2, respectively. Clearly, P represents $W_1 \bigcup W_2$. On the other hand, we add the axiom:
$P_1x \to P_2x \to Px$ meaning Px represents $W_1 \bigcap W_2$.

Problem 3

We assume W_1, W_2 are both solvable over K. Hence, all of $W_1, \widetilde{W_1}, W_2, \widetilde{W_2}$ are representable over K. Now, by Problem 2, $W_1 \bigcup W_2$ and $W_1 \bigcap W_2$ are representable over K. Also, $\widetilde{W_1} \bigcap \widetilde{W_2}$ and $\widetilde{W_1} \bigcup \widetilde{W_2}$ are both representable over K, that is, both $\widetilde{W_1 \bigcup W_2}$ and $\widetilde{W_1 \bigcap W_2}$ are representable over K, thus proving that both $W_1 \bigcup W_2$ and $W_1 \bigcap W_2$ are solvable over K.

Problem 4

(a):
This problem really asks us to find what it means for x to be the successor of y in dyadic notation. Well, the base cases are simple: $S1, 2; S2, 11$. Then, it is obvious that the successor of some dyadic representation of the form $x1$ is

$x2$. Lastly, we prove that the successor of some $x2$ is $y1$ for y the successor of x. Well, if $x2$ is some dyadic representation, and y is the successor of x, then $y1$ is just the concatenation of y and 1, that is, $x+1$ and 1, which is equal to $(x+1) \times 2^1 + 1 = 2x + 3$, while $x2$ is $2x + 2$, thus proving $Sx2, y1$ for Sx, y. Hence, the EFS over D with the axioms schemes:
$S1, 2$
$S2, 11$
$Sx1, x2$
$Sx, y \to Sx2, y1$
is such that Sx, y dyadically enumerates the relation x is the successor of y (as can easily be shown by induction).
(b):
Consider the following EFS over D (and Lx, y is to be read as x is less than y):
$Sx, y \to Lx, y$
$Lx, y \to Ly, z \to Lx, z$
Well, it is obvious that any Lx, y provable in the system is such that x is really less than y. Conversley, assume that $x < y$. Thus $x < z$ for Sx, z. Then, we can use the same rule to know that Lz, v for Sv, z, and thus we can conclude that Lx, v. And we can then repeat this process on v, and so on (that is, we show $Sv, l \to Lx, l$ and, in general, if we have some Lx, n, we use the second axiom scheme to conclude that Lx, k for Sn, k), until we finally reach y (and we will at some point), which lets us prove Lx, y, thus completing the proof.
(c):
Consider the following axiom scheme:
$Sx, y \to Sz, y \to x = z$
Clearly, every $x = z$ provable in the system is such that x really equals z. Conversley, assume $x = z$. Hence, by definition, $Sx, y \wedge Sz, y$, thus meaning the axiom will allow us to conclude that $x = z$.
(d):
Take the two axiom schemes:
$x < y \to x \leq y$
$x = y \to x \leq y$
Any $x \leq y$ provable in the system is obviously such that x really $\leq y$. Conversley, assume that $x \leq y$. Thus, either $x = y$ or $x < y$. Whatever case it is, we can use the corresponding axiom to conclude that $x \leq y$.
(e):
Take the axiom schemes:
$x < y \to x \neq y$
$x \neq y \to y \neq x$
Again, any $x \neq y$ in the systerm obviously is such that x really is not equal to y. Conversley, assume that $x \neq y$. Hence, either $x < y$ or $y < x$. In the first case, we would immediately conclude $x \neq y$. In the second, we would conclude $y \neq x$ from the first axiom, and the second axiom would let us conclude that $x \neq y$.
(f):
Take the following EFS:

$Sx, y \to x + 1 = y$
$x + y = z \to Sy, v \to Sz, w \to x + v = w$
Clearly, any $x + y = z$ provable in the system is such that $x + y$ really equals z. On the other hand, assume that $x + y = z$. Hence, $x + 1 = v$ for Sx, v is provable. Then, we can use the second axiom $y - 1$ times to conclude that $x + y = x + 1 + y - 1 = x + y = z$.
(g):
Consider the following EFS over D that has a predicate representing $x \times y = z$:
$x \times 1 = x$
$x \times y = z \to Sy, w \to z + x = v \to x \times w = v$
Again, every $x \times y = z$ provable in this system clearly is such that $x \times y = z$. Now, assume that $x \times y = z$. Thus, $x \times 1 = x$ is provable. Then, we apply the second axiom $y - 1$ times to recieve $x \times y - 1 + x = x \times y = z$
(h):
We consider the following EFS:
$x^1 = x$
$x^y = z \to Sy, v \to x^v = z \times x$
Every $x^y = z$ provable in the system obviously really means $x^y = z$. On the other hand, assume that $x^y = z$. Thus, $x^1 = x$ is provable. Then, we can apply the second axiom $y - 1$ times to receive $x^y = x \times x^{y-1} = x^y = z$.

Problem 5

We consider some dyadically enumerable functional (single-valued) relation R(x,y). We are to show that it is also dyadically solvable. Well, we consider the EFS E under which $R(x, y)$ is representable by some predicate P. Well, we have already shown the relation $x \neq y$ to be dyadically enumerable, and therefore, by Problem 1, we consider the EFS E_1 under which both $R(x, y), x \neq y$ are representable (and $R(x, y)$ is representable by Px, y). Well, we add the axiom:
$Px, y \to z \neq y \to Nx, y$
Well, because we know that Px, y does not hold for any number other than y, Nx, y represents $\sim R(x, y)$, and hence proves that $R(x, y)$ is dyadically solvable. The proof for trinary relations is similar, replacing $R(x, y)$ by $R(x, y, z)$.

Problem 6

(a), (c), (f), (g) and (h) are all dyadically solvable by Problem 5 (as they are all functional relations). (b) is dyadically solvable, as $\sim(x < y)$ can be deterimined by the axioms:
$x = y \to \sim(x < y)$

$y < x \rightarrow \sim(x < y)$, and these can be used as the other relations used in these axioms have been shown to be dyadically enumerable. (d) is dyadically solvable, as $\sim(x \leq y)$ can be determined by:
$y < x \rightarrow \sim(x \leq y)$
(e) is dyadically solvable, as $\sim(x \neq y)$ can be given by $x = y$

Problem 7

If $x \in G_n$, it follows that x is either some g_i $i < n$ or is a concatenation of strings already in the set; in either case, 1 must begin x. Also, 2 must end x in a similar fashion. Then, at no point can $11Px$, as x is a concatenation of g_i's. Lastly, it is necessary that for x to be in G_n, g_{n+1} is not a part of x (and this implies $\sim(g_m Px)$ $m > n$). Hence, we have the following Σ_0 formula expressing G_n:
$x \in G_n$ iff $1Bx \wedge 2Ex \wedge \sim(11Px) \wedge \sim(g_{n+1}Px)$.

Problem 8

Consider the following Σ_0 formula:
$(\exists v_1 \leq x)...(\exists v_k \leq x)(G_n(v_1) \wedge ... \wedge G_n(v_k) \wedge x = x')$
where k is the number of variables found in the string made up of symbols of L and variables $x_1, ..., x_k$ that defines the set $I(X)$, n is from the sub-alphabet $k_1, ..., k_n$, and x' is the concatenation resulting from, in the formula defining $I(X)$, replacing all the occurrences of symbols of L by their respective Gödel numbers, and replacing all occurrences of variables $x_1, ..., x_k$ by the variables $v_1, ..., v_k$ defined in the above formula, respectively. It is clear that, for any given set $I(X)$, the above formula holds iff $x \in I_0(X)$.

Problem 9

(a):
In general, for a given L with $m - 1$ symbols, x is a sequence number of *some* sequence iff $G_m(x) \wedge g_m Bx \wedge g_m Ex \wedge \sim g_m * g_m Px$
For suppose the formula holds. Then x is a sequence number for x, as the g_m's in x are correctly spaced, and for any term other than g_m in x, they must have a Gödel number less than g_m and are hence *some* member of L. Conversley, assume that x is a sequence number for L. By defintion, $g_m Bx \wedge g_m Ex \wedge \sim g_m * g_m Px$, Also, if $G_m(x)$ did not hold, then there would be some g_n $n > m$ in x,

which is impossible, as there are only $m-1$ terms in L. A more specific, and completely useless (as it is a restatement of the definition) formula for x being a sequence number of L for some sequence $g_m * X_1 * g_m * ... * g_m * X_r * g_m$ is $x = g_m * a_1 * g_m * ... * g_m * a_r * g_m$.
(b):
$x \in y$ iff $Seq(y) \wedge xPy \wedge \sim(g_m Px)$ where $m = n+1$ for n the number of symbols in L. *Note* : Smullyan's answer in the book is just another way of saying the above, as it is easy to see that the formula with $G_n(x)$ is equivalent to the one above.
(c):
$pr(x, y, z)$ iff $y \in z \wedge (\exists w \leq z)(wBz \wedge x \in W \wedge \sim(y \in w))$

Problem 10

Let us recall that we are considering some L in which the implication sign is part of the ordered alphabet, and therefore has a Gödel number. Thus, let e be the Gödel number of the implication sign. Hence, we have the following formula:
$Der(x, y, z)$ iff $y = xez \wedge \sim(ePx)$ and this last condition is necessary as X must be atomic.

Problem 11

(a):
$Pf(x)$ iff $Seq(x) \wedge (\forall y \leq x)[y \in x \supset$
$((\exists v \leq x)(\exists w \leq x)(Der(v, w, x) \wedge (pr(w, y, x) \wedge (pr(v, w, x)))) \vee A(y))])$ where $A(y)$ is the property of y being the Gödel number of an instance of an axiom of (E); and this is Σ_0 by the Substitution Lemma.
(b):
$ypfx$ iff:
$Pf(y) \wedge x \in y \wedge (\forall z \leq y)(\sim pr(z, x, y))$
Alternatively (as Smullyan gives in his answer), the last condition of the above formula is equivalent to $g_m * x * g_m Ey$.
(c):
$Pr(x)$ iff $\exists y(ypfx)$
(d):
We consider some formally representable set W_0 of Gödel numbers of the representable set W. So let us consider the predicate H that represetns W_0, and let its Gödel number be h. From this, we have the followng Σ_1 formula expressing W_0:
$\exists y(ypfhx)$
For n-ary relations, the above formula can be rewritten as:

$\exists y(ypfha_1c...ca_n)$ where c is the Gödel number of the comma.

Problem 12

Suppose T was formally representable. Hence, by Problem 11 (d), it would be expressed by some Σ_1 formula, thus meaning that it is arithmetic, contrary to Tarski's Theorem.

Problem 13

Suppose S is a formal arithmetic system that is ω-consistent and is Σ_0 complete. Well, this means P_0 is Σ_1, and this means P^* ($P_0^{\#}$) is also Σ_1 (by Problem 12 of Chapter 11), that is, of the form $\exists z R(x,z)$ for some Σ_0 relation $R(x,z)$. Well, because the system is Σ_0 complete, there is some formula $F(x,z)$ that is provable if R holds, and is refutable otherwise. Hence, if $x \in P^*$, $F(x,z)$ holds for at least one z, and if $x \notin P^*$, $F(x,z)$ is refutable for every z. Thus, $F(x,z)$ enumerates P^* (and this result can be generalized for any set). Then, if S is a formal system, P is formally representable. Well, consider the (E) in which P is represented by some predicate P. Then, add the axiom:
$Pnx \to Rx$ for n the Gödel number of the negation, and for some new predicate R. Hence R represents R, and therefore R is also formally representable. Hence both P^* and R^* are enumerable. Then the rest of the problem follows from identical reasoning as to that in Chapter 10.